Sanica Vinayak Myana

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Java Backend Engineer | Payments & Fraud Detection | Java 21, PostgreSQL, Docker| TDD | 2 Accepted Publications

Technical Skills

- **Languages:** Java 17/21, SQL,Python, Solidity
- **Backend:** Spring Boot, REST APIs, Microservices
- **Data & Infra:** PostgreSQL, Redis, MySQL, Docker, Kubernetes (basics), GitHub Actions CI, Linux
- **Testing:** JUnit 5
- **Risk & Security:** Anomaly detection (Isolation Forest, ONNX), risk scoring, RBAC, SIEM (Wazuh)

Experience

Systems Engineer Trainee, Infosys, Mysore — Sept 2026 – Present

- Undergoing Infosys's structured engineering training at the Mysore campus, covering Java, databases and software engineering practices ahead of project deployment.

Java Backend Intern, Coding Granth, Pune — Jan 2025 – Jul 2025

- Built and maintained backend services in Java and Spring Boot during a 6-month in-office internship, covering request validation, persistence and error handling.
- Enabled reliable client–server communication by designing and integrating RESTful APIs consumed by the team's web front end.

Security Intern, TheTechProtein — Aug 2026 – Oct 2026

- Centralised log monitoring and threat visibility across simulated enterprise networks by deploying Wazuh SIEM, and hardened Windows Active Directory (GPOs, trusts, RODC) in Linux/Windows Server labs.

Projects

Multi-Currency Payments Ledger | *Java 21, Javalin, jOOQ, PostgreSQL, Redis, Spock, Docker, GCP* | GitHub
Nov 2026

- Guaranteed zero balance drift across 10,000 concurrent transfers, verified by Spock concurrency tests, by modelling money movement as double-entry journal entries with row-level locking in PostgreSQL via jOOQ.
- Eliminated duplicate charges on client retries (100% of replayed requests deduplicated) by implementing idempotency keys in Redis, with PostgreSQL as the source of truth.
- Sustained 800+ req/s at p99 latency under 50 ms in Gatling load tests by tuning connection pooling and composite indexes; shipped through GitHub Actions CI and Docker to GCP Cloud Run.
- Reached 90% test coverage by building test-first with Spock and Testcontainers against a real PostgreSQL instance.

Real-Time Fraud Risk Scoring Service | *Java 21, Redis Streams, ONNX Runtime, PostgreSQL, Docker* | GitHub
Dec 2026

- Scored payment events in under 15 ms p99 by combining Redis sliding-window velocity rules with an Isolation Forest model served in-JVM through ONNX Runtime.
- Cut false positives by 25% against a rules-only baseline on 100,000 synthetic transactions by blending ML anomaly scores and rule signals into one explainable risk score with reason codes.
- Enabled zero-risk rule rollouts by adding a shadow mode that logs decisions without blocking, consuming transfer events from the ledger service over Redis Streams.

UEBA: User & Entity Behaviour Analytics | *Python, Isolation Forest, Risk Scoring* | 1st Prize, B.E. Project Competition
Sept 2025

- Cut false positives by 20% by engineering behavioural features from raw security logs and training an Isolation Forest anomaly detection model.
- Sped up incident response by 30% by feeding real-time anomaly scores and risk assessments into a security operations workflow; basis of an accepted Scopus-indexed publication.

Blockchain-Based E-Voting Platform | *Solidity, Ethers.js, MetaMask, Ethereum* | GitHub — Feb 2024

- Ran tamper-proof college club elections for 50+ voters by designing Solidity smart contracts that record each vote on-chain, with a web front end connected through Ethers.js and MetaMask.
- Basis of an accepted IEEE paper on privacy-preserving voting using temporal decorrelation in Ethereum systems.

Publications

- *Adaptive Behavioral Anomaly Detection: Integrating UEBA and EDR for Real-Time Mitigation of Threats and Insider Risks.* Scopus-indexed journal, 2026 (accepted)
- *Privacy-Preserving Blockchain-Based Voting Using Temporal Decorrelation in Ethereum Systems.* IEEE, 2026 (accepted); smart-contract voting system piloted with 50+ voters.

Education

B.E. in Computer Engineering, Marathwada Mitra Mandal's College of Engineering (SPPU), Pune 2022 – 2026
CGPA: 8.81/10 (First Class with Distinction) | *Coursework:* Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks

Achievements & Leadership

- 1st Prize, B.E. Project Competition, MMCOE (UEBA) | Top 3 at Avishkar (college level), selected for SPPU-level representation
- Led a Java workshop for women engineering students at PICT, Pune